

Docker

Conteneurisation — de hello-world à la stack LAMP

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Bachelor IT Cybersécurité — La Plateforme

`Docker Engine · Compose · Registry · Portainer` `11 jobs + bonus` `Debian 13`

Job 01 — Installer Docker

- Debian 13 (VMware), dépôt officiel Docker
- Paquets docker-ce, CLI, containerd, buildx, compose
- Utilisateur ajouté au groupe docker
- Vérification : service actif

```
$ sudo apt install docker-ce docker-ce-cli
containerd.io
$ sudo usermod -aG docker $USER
$ systemctl status docker
```

```
root@debian:~# systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; pres>
     Active: active (running) since Wed 2026-10-07 11:07:49 CEST; 10s ago
 Invocation: 758b8340812049e489c5391b28c1d0d8
TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
   Main PID: 2318 (dockerd)
      Tasks: 8
     Memory: 29.7M (peak: 31M)
        CPU: 373ms
     CGroup: /system.slice/docker.service
             └─2318 /usr/bin/dockerd -H fd:// --containerd=/run/containerd>
```

⚡ PROBLÈME RÉSOLU

Symptôme Dépôt cdrom bloquant apt **Solution** Ligne cdrom du sources.list commentée avant apt update

Job 02 — Premier conteneur

- Image hello-world tirée du Docker Hub
- docker run = pull + create + start
- Voir, arrêter, supprimer le conteneur
- Différence image / conteneur

```
$ docker run hello-world
$ docker ps -a
$ docker rm <id> ; docker rmi hello-world
```

```
root@debian:~# docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5e230903533324d887c48ad5e5c56d294eab81588df9605b07d1afe895f9cc8f8
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

root@debian:~#
```

Job 03 — Image depuis un Dockerfile

- Dockerfile : FROM debian, COPY message.txt, CMD cat
- docker build -t hello-debian .
- Le conteneur affiche le fichier
- diff : sortie identique à message.txt

```
$ docker build -t hello-debian .
$ docker run --rm hello-debian
$ docker run --rm hello-debian | diff - message.txt
```

```
mkdir -p ~/docker/job03 && cd ~/docker/job03
Untagged: mon-hello:latest
Deleted: sha256:8b9de08b4939d6a143025af43a2dff446484ba0320117077cdd452d08ba4430a
alexis@debian:~/docker/job03$ docker run --rm hello-world > message.txt
alexis@debian:~/docker/job03$ cat > Dockerfile <<'EOF'
FROM debian:stable-slim
COPY message.txt /message.txt
CMD ["cat", "/message.txt"]
EOF
alexis@debian:~/docker/job03$ docker build -t hello-debian .
docker run --rm hello-debian
docker run --rm hello-world
[+] Building 4.7s (7/7) FINISHED                                docker:default
 ⇒ [internal] load build definition from Dockerfile                      0.0s
 ⇒ ⇒ transferring dockerfile: 119B                                       0.0s
 ⇒ [internal] load metadata for docker.io/library/debian:stable-sli      0.8s
 ⇒ [internal] load .dockerignore                                         0.0s
 ⇒ ⇒ transferring context: 2B                                            0.0s
 ⇒ [internal] load build context                                         0.0s
 ⇒ ⇒ transferring context: 847B                                          0.0s
 ⇒ [1/2] FROM docker.io/library/debian:stable-slim@sha256:eb593cf2c      1.7s
 ⇒ ⇒ resolve docker.io/library/debian:stable-slim@sha256:eb593cf2c       0.0s
 ⇒ ⇒ extracting sha256:90ecb15ebb389405c9fa3c6d01d1f22d7be80cbb79a       1.7s
 ⇒ [2/2] COPY message.txt /message.txt                                   0.1s
 ⇒ exporting to image                                                    2.0s
 ⇒ ⇒ exporting layers                                                    0.0s
 ⇒ ⇒ exporting manifest sha256:bdf00715a5ce0104827a7ddac8ea3fe3c57       0.0s
 ⇒ ⇒ exporting config sha256:110fee6547090288c64bb1057c56c8b807a93       0.0s
 ⇒ ⇒ exporting attestation manifest sha256:49c2e663afb762683d6f2a9       0.0s
 ⇒ ⇒ exporting manifest list sha256:6f15d3bb4a4000176cff734e16ac0a       0.0s
 ⇒ ⇒ naming to docker.io/library/hello-debian:latest                     0.0s
 ⇒ ⇒ unpacking to docker.io/library/hello-debian:latest                  1.8s
```

Job 03 — Image depuis un Dockerfile (suite)

```
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    executable that produces the output you are currently reading.
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alexis@debian:~/docker/job03$ diff <(docker run --rm hello-debian) <(docker run --rm hello-world) && echo "Identique"
Identique
alexis@debian:~/docker/job03$
```

Job 04 — Conteneur SSH

- Dockerfile : debian + openssh-server
- Utilisateur créé, EXPOSE 22
- Port publié : -p 2222:22
- Connexion ssh depuis la VM

```
$ docker build -t ssh-debian .
$ docker run -d -p 2222:22 --name ssh-test ssh-debian
$ ssh -p 2222 user@localhost
```

```
alexis@debian:~/docker/job04$ docker build -t ssh-debian .
docker run -d --name ssh-test -p 2222:22 ssh-debian
docker ps
[+] Building 14.8s (6/6) FINISHED                          docker:default
 => [internal] load build definition from Dockerfile                  0.0s
 => => transferring dockerfile: 379B                                  0.0s
 => [internal] load metadata for docker.io/library/debian:stable-sli  1.0s
 => [internal] load .dockerignore                                     0.0s
 => => transferring context: 2B                                       0.0s
 => CACHED [1/2] FROM docker.io/library/debian:stable-slim@sha256:eb  0.0s
 => => resolve docker.io/library/debian:stable-slim@sha256:eb593cf2c  0.0s
 => [2/2] RUN apt-get update &&     apt-get install -y --no-install  12.0s
 => exporting to image                                                1.7s
 => => exporting layers                                               1.3s
 => => exporting manifest sha256:fc77d264021092e065c7b646839a004206a  0.0s
 => => exporting config sha256:287e86b66e3cfb48d0b3a5eb8c2963658c610  0.0s
 => => exporting attestation manifest sha256:17e59c73ffb5f093241c650  0.0s
 => => exporting manifest list sha256:3ed4ff445c550e65cf596015d5ad96  0.0s
 => => naming to docker.io/library/ssh-debian:latest                  0.0s
 => => unpacking to docker.io/library/ssh-debian:latest               0.3s
a372aaa732f9a56ca2dec637b08ccec96be2a141d28319f97601aa8def4616da
CONTAINER ID   IMAGE        COMMAND               CREATED                  STATUS                  PORTS                                       NAMES
a372aaa732f9   ssh-debian   "/usr/sbin/sshd -D"   Less than a second ago   Up Less than a second   0.0.0.0:2222→22/tcp, [::]:2222→22/tcp   ssh-test
alexis@debian:~/docker/job04$ ssh root@IP_VM -p 2222
ssh: Could not resolve hostname ip_vm: Name or service not known
alexis@debian:~/docker/job04$ ssh root@192.168.109.135 -p 2222
The authenticity of host '[192.168.109.135]:2222 ([192.168.109.135]:2222)' can't be established.
ED25519 key fingerprint is SHA256:nNn01DzyvPkptRDlSwq7bFEG3AsBwgTRJjz3TqS4Us0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[192.168.109.135]:2222' (ED25519) to the list of known hosts.
root@192.168.109.135's password:
Linux a372aaa732f9 6.12.111+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.111-1 (2026-09-28) x86_64

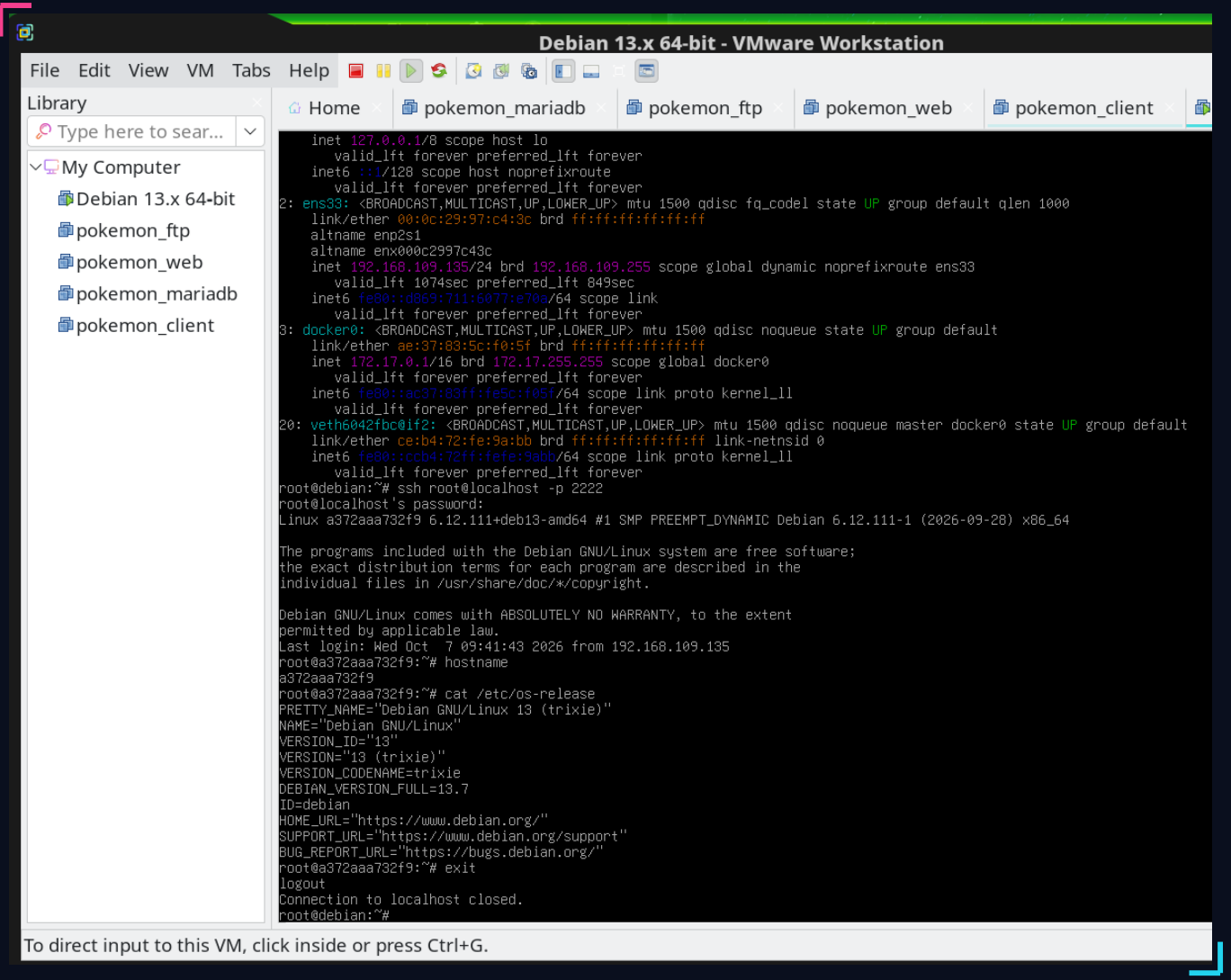
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Oct  7 09:39:40 2026 from 172.17.0.1
root@a372aaa732f9:~#
```

⚡ PROBLÈME RÉSOLU

Symptôme scp lancé depuis le conteneur, ou ssh vers « ip_vm » non résolu **Solution** Commandes exécutées depuis la VM / le PC, avec la vraie adresse IP

Job 04 — Conteneur SSH (suite)



Job 05 — Alias des commandes Docker

- Raccourcis pour les commandes CLI fréquentes
- Définis dans ~/.bashrc
- Rechargés avec source ~/.bashrc
- Gain de temps au quotidien

```
$ alias dps='docker ps'
$ alias dpa='docker ps -a'
$ source ~/.bashrc
```

```
# Alias Docker
alias d='docker'
alias dps='docker ps'
alias dpsa='docker ps -a'
alias di='docker images'
alias dr='docker run'
alias drit='docker run -it'
alias dex='docker exec -it'
alias dlogs='docker logs'
alias dstart='docker start'
alias dstop='docker stop'
alias drm='docker rm'
alias drmi='docker rmi'
alias dbuild='docker build -t'
alias dprune='docker system prune'
alias dstopall='docker stop $(docker ps -q)'
alias drmall='docker rm $(docker ps -aq)'
alias dc='docker compose'
```

```
alexis@debian:~$ alias | grep docker
dpsa
di
dex ssh-test bash
alias d='docker'
alias dbuild='docker build -t'
alias dc='docker compose'
alias dex='docker exec -it'
alias di='docker images'
alias dlogs='docker logs'
alias dprune='docker system prune'
alias dps='docker ps'
alias dpsa='docker ps -a'
alias dr='docker run'
alias drit='docker run -it'
alias drm='docker rm'
alias drmall='docker rm $(docker ps -aq)'
alias drmi='docker rmi'
alias dstart='docker start'
alias dstop='docker stop'
alias dstopall='docker stop $(docker ps -q)'
CONTAINER ID   IMAGE          COMMAND                  CREATED             STATUS                       PORTS                                       NAMES
a372aaa732f9   ssh-debian     "/usr/sbin/sshd -D"      38 minutes ago      Up 38 minutes                0.0.0.0:2222->22/tcp, [::]:2222->22/tcp     ssh-test
6a50423b145b   hello-world    "/hello"                 About an hour ago   Exited (0) About an hour ago                                             pedantic_raman
ujan

                                                                                 i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-debian:latest  6f15d3bb4a40       118MB         29.8MB
hello-world:latest   5e2309035332      25.9kB         9.49kB    U
ssh-debian:latest    3ed4ff445c55       160MB         40.6MB    U
root@a372aaa732f9:/#
```

Job 06 — Volumes

- Volume nommé mon-volume
- Monté dans deux conteneurs c1 et c2
- Ce que c1 écrit, c2 le lit
- Les données survivent à la suppression du conteneur

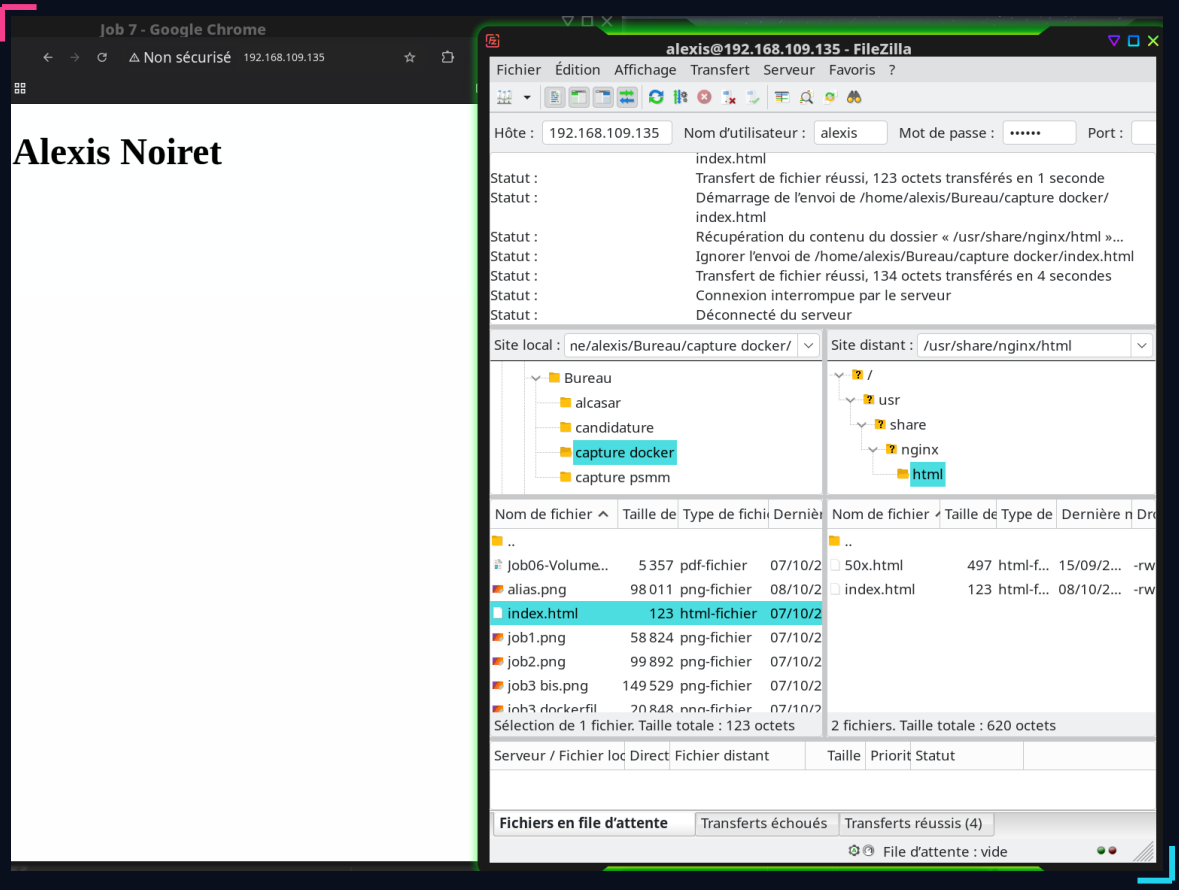
```
$ docker volume create mon-volume
$ docker run -d --name c1 -v mon-volume:/data alpine
sleep 3600
$ docker exec c2 cat /data/test.txt
```

```
alexis@debian:~$ docker volume create mon-volume
mon-volume
alexis@debian:~$ docker volume ls
DRIVER    VOLUME NAME
local     mon-volume
alexis@debian:~$ docker volume inspect mon-volume
[
    {
        "CreatedAt": "2026-10-07T13:24:48+02:00",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/mon-volume/_data",
        "Name": "mon-volume",
        "Options": null,
        "Scope": "local"
    }
]
alexis@debian:~$ docker run --rm -v mon-volume:/data debian bash -c "echo 'bonjour' > /data/test.txt"
Unable to find image 'debian:latest' locally
latest: Pulling from library/debian
4846f2900730: Pull complete
50f5607b0ccc: Download complete
Digest: sha256:913f6706df59a68922d1dd08f78c2476560a8d367897200a6005b00e5f67c2d5
Status: Downloaded newer image for debian:latest
alexis@debian:~$ docker run --rm -v mon-volume:/data debian cat /data/test.txt
bonjour
alexis@debian:~$ docker run -d --name c1 -v mon-volume:/data debian sleep 3600
03f2e29117e202aa5bcd312f48161324b6c27b7fed653cdfa53e463180f02ce5
alexis@debian:~$ docker run -d --name c2 -v mon-volume:/data debian sleep 3600
d86a5b2650b508e322bb176fca37bf1da828b5c33b085b04ad270a2f1bed10a7
alexis@debian:~$ docker exec c1 bash -c "echo 'depuis c1' >> /data/test.txt"
alexis@debian:~$ docker exec c2 cat /data/test.txt
bonjour
depuis c1
alexis@debian:~$ docker rm -f c1 c2
docker volume rm mon-volume
docker volume prune
c1
c2
mon-volume
WARNING! This will remove anonymous local volumes not used by at least one container.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
alexis@debian:~$
```

Job 07 — Nginx + FTP, volume partagé

- docker compose : web (nginx) + ftp (alpine-ftp-server)
- Même volume monté dans les deux services
- Mode passif FTP : ports 21000-21010
- Dépôt de index.html avec FileZilla → page mise à jour

```
$ docker compose up -d
$ docker exec web-ftp chown -R alexis /var/www/html
$ FileZilla → ftp://alexis@192.168.109.135
```



⚡ PROBLÈME RÉSOLU

Symptôme Dépôt FTP refusé (permissions) **Solution** chown du dossier partagé dans le conteneur FTP

Job 08 — Image nginx personnalisée

- FROM debian + apt install nginx + index.html
- Image nginx-debian vs image officielle nginx
- Port 8081 → 80
- Comparaison de taille des images

```
$ docker build -t nginx-debian .
$ docker run -d -p 8081:80 --name nginx-test nginx-debian
$ curl http://localhost:8081
```

```
alexis@debian:~/docker/job08$ docker build -t nginx-debian .
docker run -d --name nginx-test -p 8081:80 nginx-debian
docker ps
curl http://localhost:8081
[+] Building 13.9s (8/8) FINISHED                          docker:default
 ⇒ [internal] load build definition from Dockerfile                 0.0s
 ⇒ ⇒ transferring dockerfile: 266B                                  0.0s
 ⇒ [internal] load metadata for docker.io/library/debian:stable     1.2s
 ⇒ [internal] load .dockerignore                                    0.0s
 ⇒ ⇒ transferring context: 2B                                       0.0s
 ⇒ CACHED [1/3] FROM docker.io/library/debian:stable-slim@sha25     0.0s
 ⇒ ⇒ resolve docker.io/library/debian:stable-slim@sha256:eb593      0.0s
 ⇒ [internal] load build context                                    0.1s
 ⇒ ⇒ transferring context: 191B                                     0.0s
 ⇒ [2/3] RUN apt-get update &&     apt-get install -y --no-ins     11.5s
 ⇒ [3/3] COPY index.html /var/www/html/index.html                   0.1s
 ⇒ exporting to image                                               0.9s
 ⇒ ⇒ exporting layers                                               0.7s
 ⇒ ⇒ exporting manifest sha256:3191e433d8e506969fe1f7b34aedd47      0.0s
 ⇒ ⇒ exporting config sha256:91428876b6190f2bc672c6665d254f54e      0.0s
 ⇒ ⇒ exporting attestation manifest sha256:06abce2e5d77cd8ab46      0.0s
 ⇒ ⇒ exporting manifest list sha256:c904c5ea33534cafd7d4bbd168      0.0s
 ⇒ ⇒ naming to docker.io/library/nginx-debian:latest                0.0s
 ⇒ ⇒ unpacking to docker.io/library/nginx-debian:latest             0.2s
009c72e819c6ee45fc156e071b0331c6c621d03efb5d6a70eb305c9af9f9d271
CONTAINER ID   IMAGE                     COMMAND                  CREATED                  STATUS                  PORTS                                                
```

Job 08 — Image nginx personnalisée (suite)




```
alexis@debian:~/docker/job08$ cat Dockerfile
docker images
FROM debian:stable-slim

RUN apt-get update && \
    apt-get install -y --no-install-recommends nginx && \
    rm -rf /var/lib/apt/lists/*

COPY index.html /var/www/html/index.html

EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
                                        i Info →  U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
debian:latest        913f6706df59       187MB         52.6MB
delfer/alpine-ftp-server:latest
                     60bb774d8408      13.7MB         3.86MB    U
hello-debian:latest  6f15d3bb4a40       118MB         29.8MB
hello-world:latest   5e2309035332      25.9kB         9.49kB    U
nginx-debian:latest  c904c5ea3353       132MB         33.6MB    U
nginx:latest         f9ea18bfa4fa       243MB         66.7MB    U
ssh-debian:latest    3ed4ff445c55       160MB         40.6MB    U
alexis@debian:~/docker/job08$ docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED             STATUS              PORTS
                                                                 NAMES
009c72e819c6   nginx-debian              "nginx -g 'daemon of…"   About a minute ago  Up About a minute   0.0.0.0:8081→80/tcp, [::]:8081→80/tcp
                                                                 nginx-test
269e90d7d3b8   nginx:latest              "/docker-entrypoint.…"   23 hours ago        Up About an hour    0.0.0.0:80→80/tcp, [::]:80→80/tcp
                                                                 web-nginx
01258d34e3cd   delfer/alpine-ftp-server  "/sbin/tini -- /bin/…"   23 hours ago        Up About an hour    0.0.0.0:21→21/tcp, [::]:21→21/tcp, 0.0.0.0:21000
-21010→21000-21010/tcp, [::]:21000-21010→21000-21010/tcp   web-ftp
```

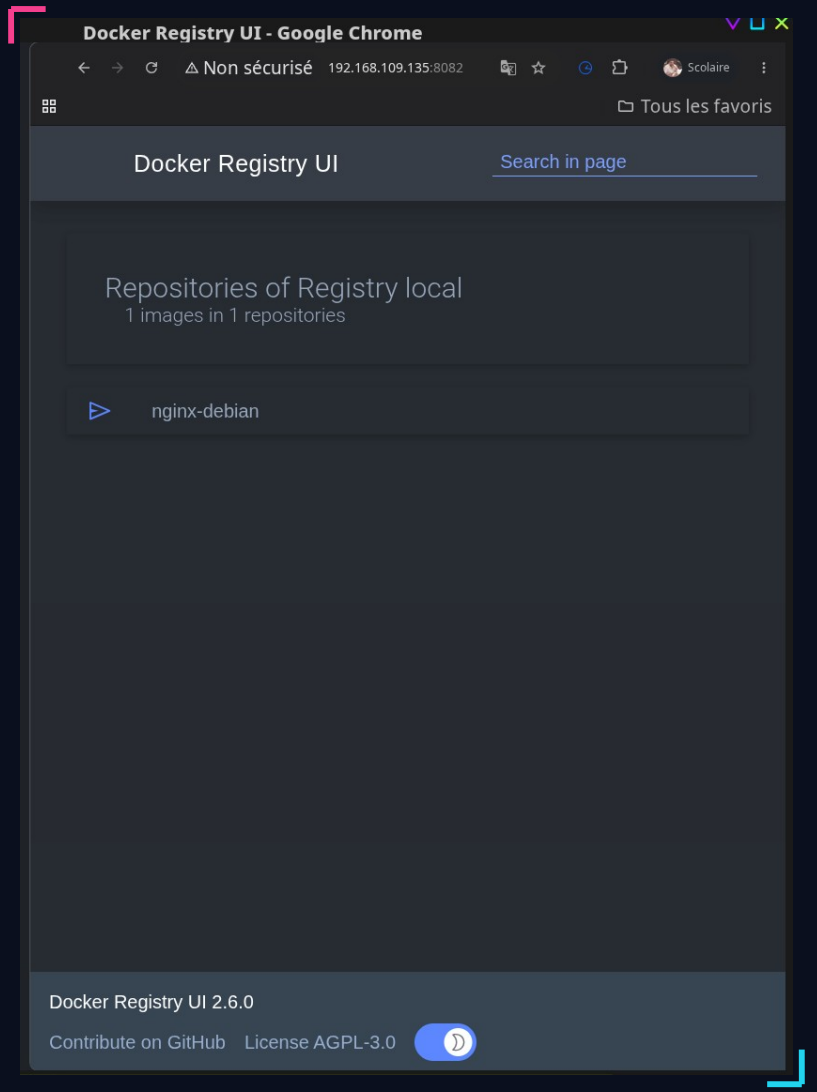
Job 09 — Registry local + interface

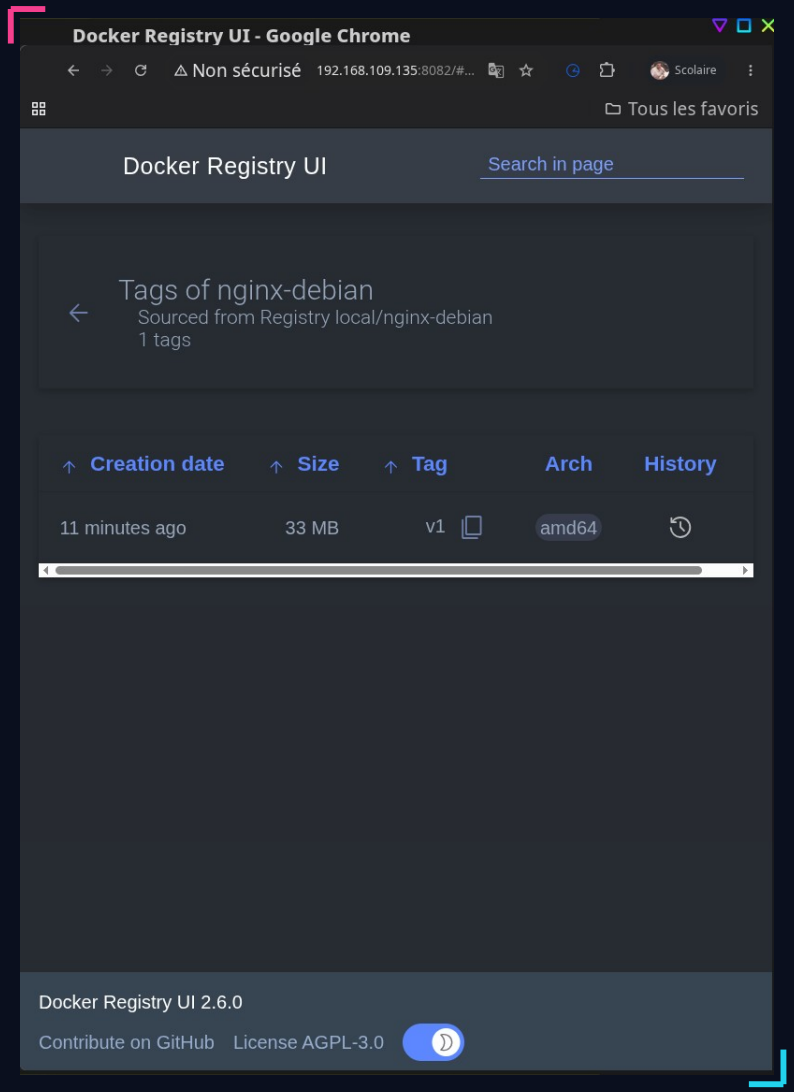
- registry:2 + joxit/docker-registry-ui
- Interface web sur le port 8082
- Tag, push, catalogue, suppression locale, pull
- localhost:5000 autorisé en non sécurisé

```
$ docker tag nginx-debian:job08 localhost:5000/nginx-debian:job08
$ docker push localhost:5000/nginx-debian:job08
$ curl http://localhost:5000/v2/_catalog
```



Job 09 — Registry local + interface (suite)





Job 10 — Scripts install et uninstall

- install-docker.sh : 5 étapes numérotées [1/5]–[5/5]
- uninstall-docker.sh : confirmation (oui/non)
- Supprime conteneurs, volumes, images, paquets, dépôt
- Retour à un système propre

```
$ sudo bash install-docker.sh
$ sudo bash uninstall-docker.sh
$ docker: command not found
```

```
update-alternatives: utilisation de « /usr/sbin/iptables-legacy » pour fournir « /usr/sbin/iptables » (iptables) en mode automatique
update-alternatives: utilisation de « /usr/sbin/ip6tables-legacy » pour fournir « /usr/sbin/ip6tables » (ip6tables) en mode automatique
update-alternatives: utilisation de « /usr/sbin/iptables-nft » pour fournir « /usr/sbin/iptables » (iptables) en mode automatique
update-alternatives: utilisation de « /usr/sbin/ip6tables-nft » pour fournir « /usr/sbin/ip6tables » (ip6tables) en mode automatique
update-alternatives: utilisation de « /usr/sbin/arptables-nft » pour fournir « /usr/sbin/arptables » (arptables) en mode automatique
update-alternatives: utilisation de « /usr/sbin/ebtables-nft » pour fournir « /usr/sbin/ebtables » (ebtables) en mode automatique
Paramétrage de docker-ce (5:29.8.2-1~debian.13~trixie) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' → '/usr/lib/systemd/system/docker.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' → '/usr/lib/systemd/system/docker.socket'.
Traitement des actions différées (« triggers ») pour man-db (2.13.1-1) ...
Traitement des actions différées (« triggers ») pour libc-bin (2.41-12+deb13u4) ...
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
[5/5] Ajout de l'utilisateur au groupe docker ...
Déconnecte-toi puis reconnecte-toi pour utiliser docker sans sudo.
Test ...
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5e23090353324d887c48ad5e5c56d294eab81588df9605b07d1afe895f9cc8f8
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
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 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
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Docker est installé.
alexis@debian:~/docker/job10$ docker --version
docker ps
docker compose version
Docker version 29.8.2, build 7fc2dff
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
Docker Compose version v5.6.0
```

⚡ PROBLÈME RÉSOLU

Symptôme apt verrouillé par un autre processus **Solution** DPkg::Lock::Timeout=120 dans la boucle de purge

Job 10 — Scripts install et uninstall (suite)

```
Après cette opération, 32,7 MB d'espace disque seront libérés.
(Lecture de la base de données ... 37661 fichiers et répertoires déjà installés.)
Suppression de docker-compose-plugin (5.6.0-1~debian.13~trixie) ...
Lecture des listes de paquets ... Fait
Construction de l'arbre des dépendances ... Fait
Lecture des informations d'état ... Fait
Les paquets suivants ont été installés automatiquement et ne sont plus nécessaires :
  git git-man iptables libcurl3t64-gnutls liberror-perl libip6tc2
  libnetfilter-conntrack3 libnfnetlink0 libngtcp2-16
  libngtcp2-crypto-gnutls8 patch pigz
Veuillez utiliser « sudo apt autoremove » pour les supprimer.
Les paquets suivants seront ENLEVÉS :
  docker-ce-rootless-extras*
0 mis à jour, 0 nouvellement installés, 1 à enlever et 2 non mis à jour.
Après cette opération, 26,1 MB d'espace disque seront libérés.
(Lecture de la base de données ... 37655 fichiers et répertoires déjà installés.)
Suppression de docker-ce-rootless-extras (5:29.8.2-1~debian.13~trixie) ...
Lecture des listes de paquets ... Fait
Construction de l'arbre des dépendances ... Fait
Lecture des informations d'état ... Fait
Les paquets suivants seront ENLEVÉS :
  git* git-man* iptables* libcurl3t64-gnutls* liberror-perl*
  libip6tc2* libnetfilter-conntrack3* libnfnetlink0* libngtcp2-16*
  libngtcp2-crypto-gnutls8* patch* pigz*
0 mis à jour, 0 nouvellement installés, 12 à enlever et 2 non mis à jour.
Après cette opération, 57,3 MB d'espace disque seront libérés.
(Lecture de la base de données ... 37650 fichiers et répertoires déjà installés.)
Suppression de git (1:2.47.3-0+deb13u1) ...
Suppression de git-man (1:2.47.3-0+deb13u1) ...
Suppression de iptables (1.8.11-2) ...
Suppression de libcurl3t64-gnutls:amd64 (8.14.1-2+deb13u5) ...
Suppression de liberror-perl (0.17030-1) ...
Suppression de libip6tc2:amd64 (1.8.11-2) ...
Suppression de libnetfilter-conntrack3:amd64 (1.1.0-1) ...
Suppression de libnfnetlink0:amd64 (1.0.2-3) ...
Suppression de libngtcp2-crypto-gnutls8:amd64 (1.11.0-1+deb13u1) ...
Suppression de libngtcp2-16:amd64 (1.11.0-1+deb13u1) ...
Suppression de patch (2.8-2) ...
Suppression de pigz (2.8-1) ...
Traitement des actions différées (« triggers ») pour man-db (2.13.1-1) ...
Traitement des actions différées (« triggers ») pour libc-bin (2.41-12+deb13u4) ...
(Lecture de la base de données ... 36246 fichiers et répertoires déjà installés.)
Purge des fichiers de configuration de git (1:2.47.3-0+deb13u1) ...
[4/5] Suppression des données et de la configuration ...
[5/5] Mise à jour de l'index apt ...
Atteint : 1 http://deb.debian.org/debian trixie InRelease
Atteint : 2 http://deb.debian.org/debian trixie-updates InRelease
Atteint : 3 http://security.debian.org/debian-security trixie-security InRelease
Lecture des listes de paquets ... Fait
Docker est complètement supprimé.
```

Job 11 — Portainer : installation

- Portainer CE : interface web de Docker
- Lui-même un conteneur, pilote docker.sock
- Compte admin créé dans les 5 minutes
- Environnement « local » à sélectionner

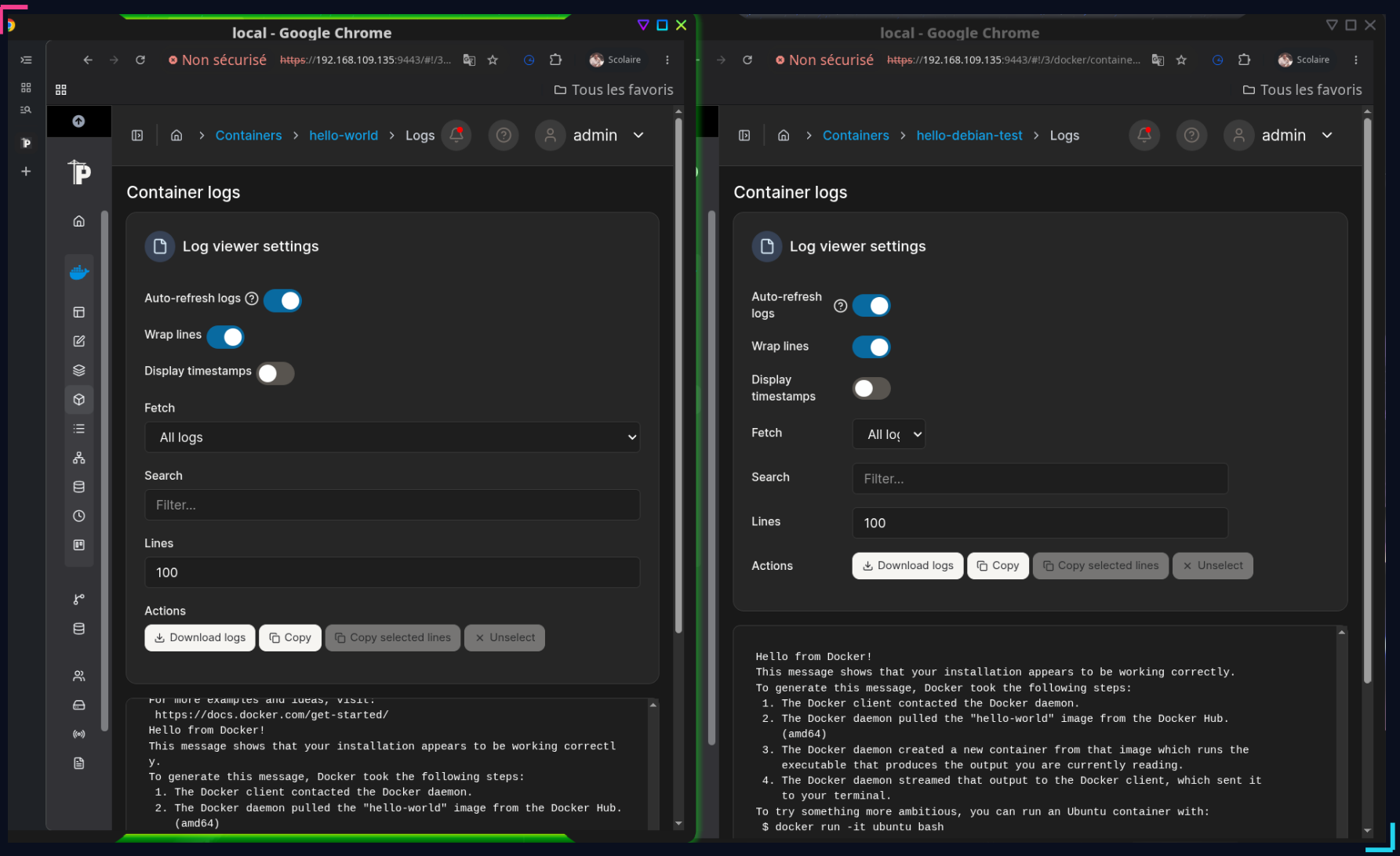
```
$ docker volume create portainer_data
$ docker run -d -p 9443:9443 --name portainer --
restart=always \
  -v /var/run/docker.sock:/var/run/docker.sock -v
portainer_data:/data portainer/portainer-ce:lts
```

```
alexis@debian:~$ docker volume create portainer_data
docker run -d --name portainer --restart=always \
  -p 9443:9443 \
  -v /var/run/docker.sock:/var/run/docker.sock \
  -v portainer_data:/data \
  portainer/portainer-ce:lts
docker ps
portainer_data
Unable to find image 'portainer/portainer-ce:lts' locally
lts: Pulling from portainer/portainer-ce
4f4fb700ef54: Pull complete
0602b0461d81: Pull complete
496a609b2162: Pull complete
70351e344504: Pull complete
6b1a001e5723: Pull complete
73d87cdfeed7: Pull complete
92468d92c28f: Pull complete
77b1cbea6f2c: Download complete
6975e4d3b2ff: Download complete
Digest: sha256:03f94a406496dc84d37b552795fc424ccad8d0d7200353b6a82915f6e8f73ab0
Status: Downloaded newer image for portainer/portainer-ce:lts
c1030c42828abf23fff7bc46ce6655d80b30d8271eefa5093b572f2fcecba68b
CONTAINER ID   IMAGE                          COMMAND        CREATED         STATUS
           PORTS                                                  NAMES
c1030c42828a   portainer/portainer-ce:lts   "/portainer"   1 second ago   Up Less than
 a second   8000/tcp, 9000/tcp, 0.0.0.0:9443→9443/tcp, [::]:9443→9443/tcp   portaine
r
alexis@debian:~$
```

⚡ PROBLÈME RÉSOLU

Symptôme docker.sock monté = accès root à l'hôte **Solution** Mot de passe fort, port 9443 non exposé

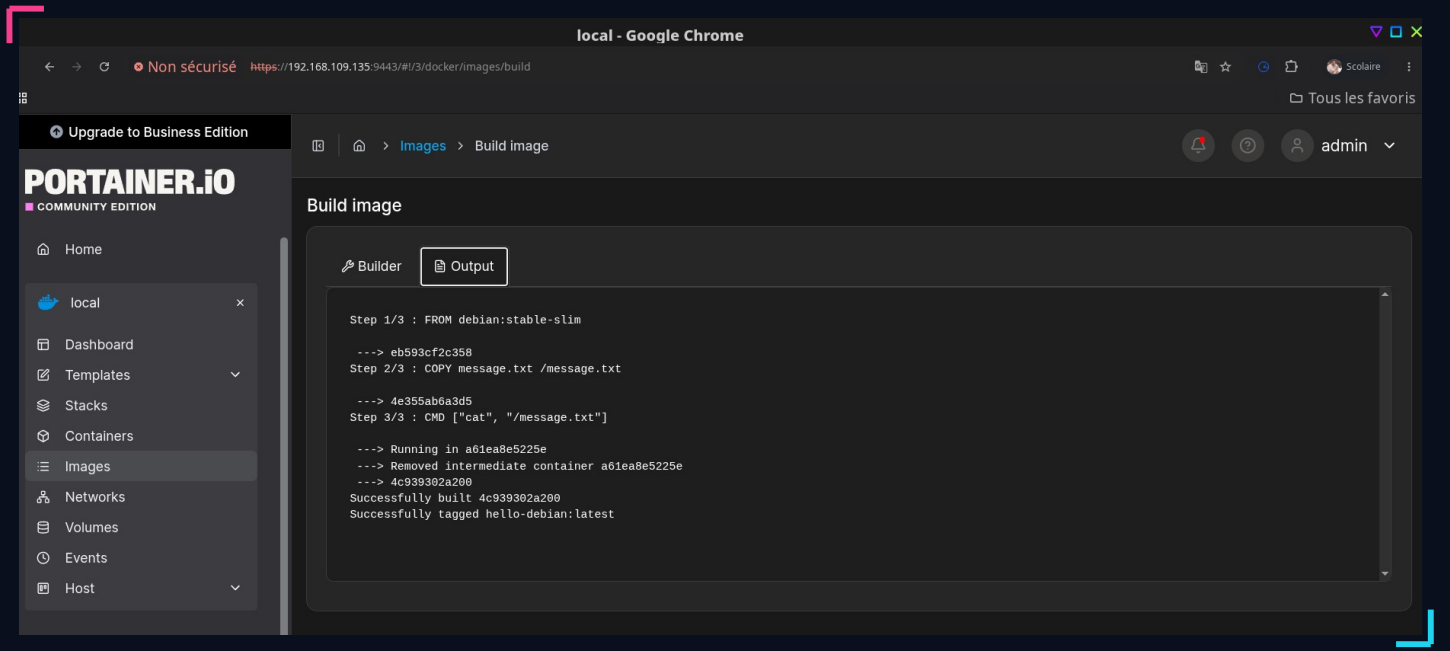
Job 11 — Portainer : installation (suite)



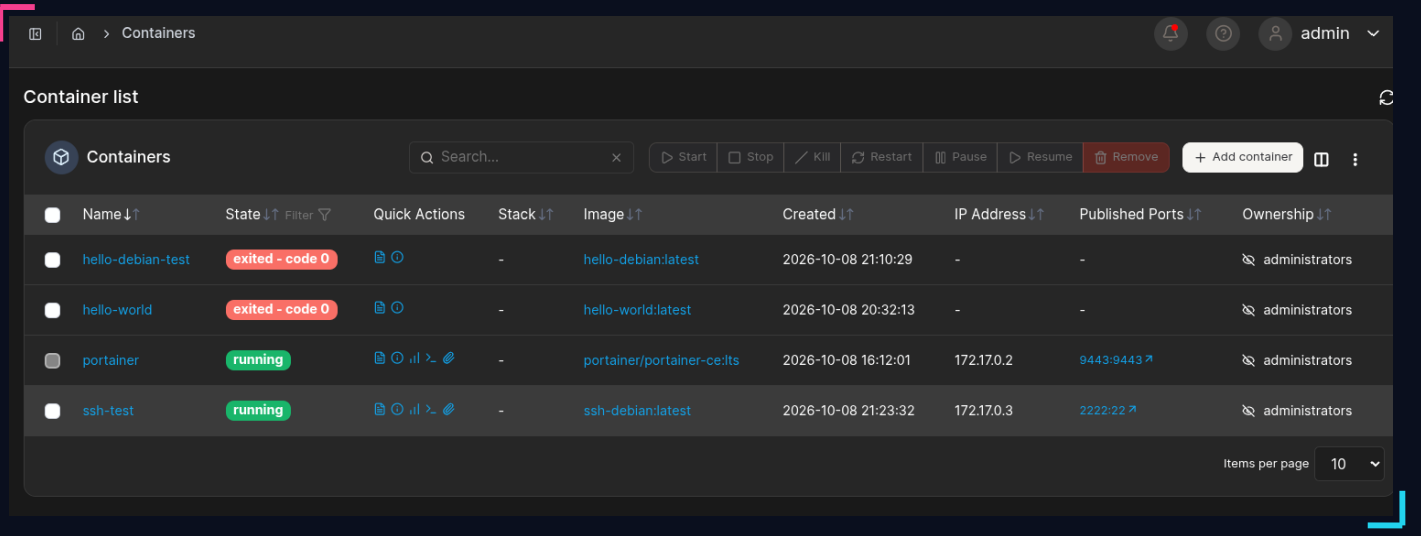
Job 11 — Refaire les jobs 3, 4 et 6

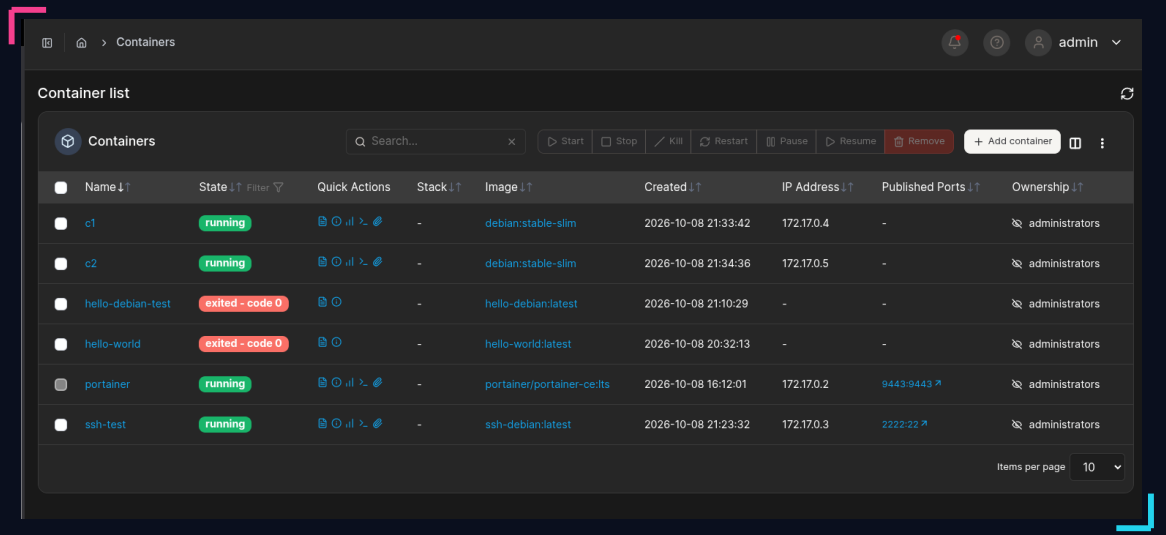
- Job 3 : build via archive .tar (Dockerfile à la racine)
- Job 4 : build en web editor, port 2222 → 22
- Job 6 : volume partagé par c1 et c2, console web

```
$ Images → Build a new image
$ Containers → Add container
$ Volumes → Add volume
```



Job 11 — Refaire les jobs 3, 4 et 6 (suite)

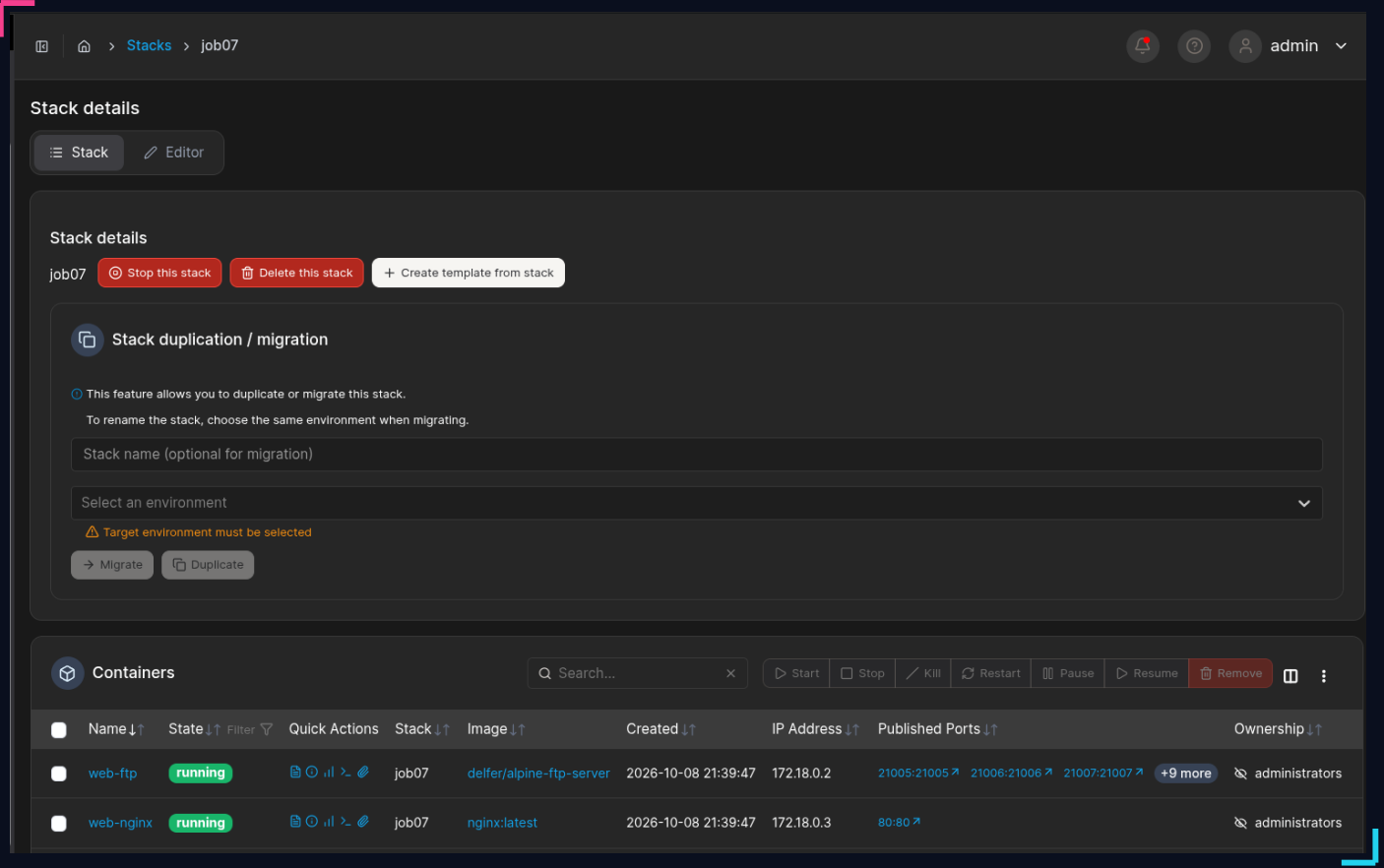




Job 11 — Refaire les jobs 7, 8 et 9

- Job 7 : stack Compose dans le web editor
- Job 8 : build + conteneur 8081 → 80
- Job 9 : stack registry + registry local déclaré
- « Always pull the image » désactivé pour les images locales

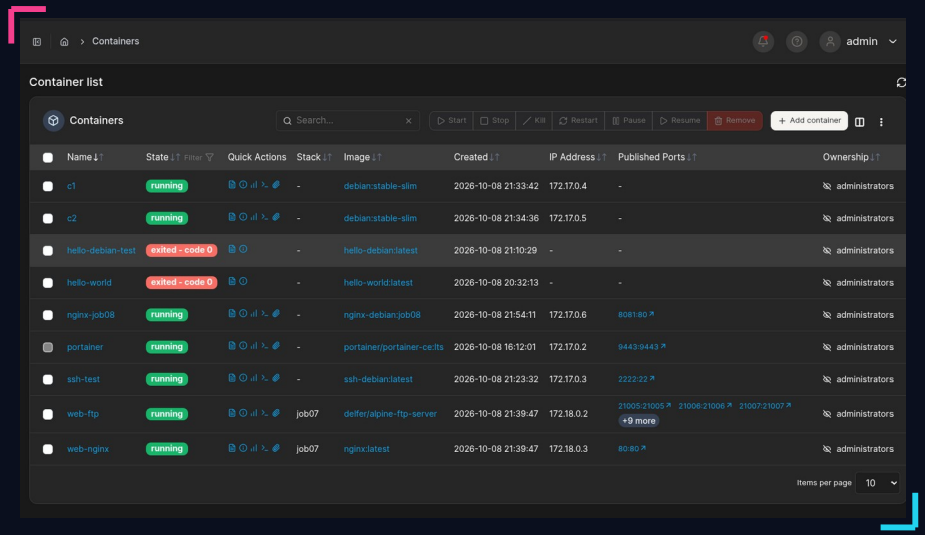
```
$ Stacks → Add stack (web editor)
$ Registries → Add registry → Custom
```

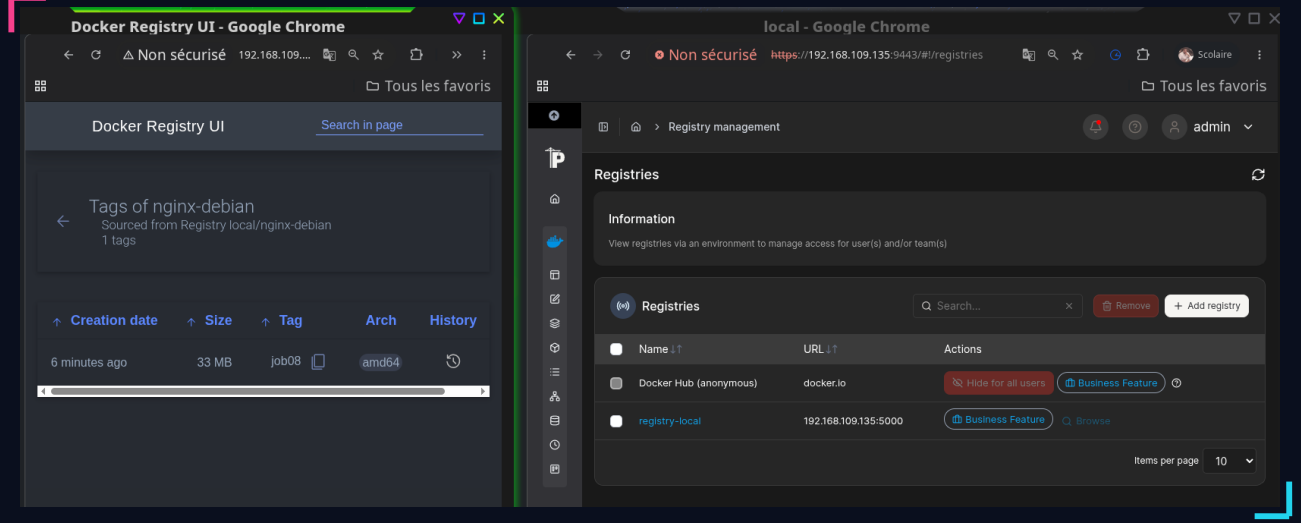


⚡ PROBLÈME RÉSOLU

Symptôme Portainer tente de tirer une image locale du Docker Hub **Solution** Décocher « Always pull the image »

Job 11 — Refaire les jobs 7, 8 et 9 (suite)





Job 11 — Alternatives à Portainer

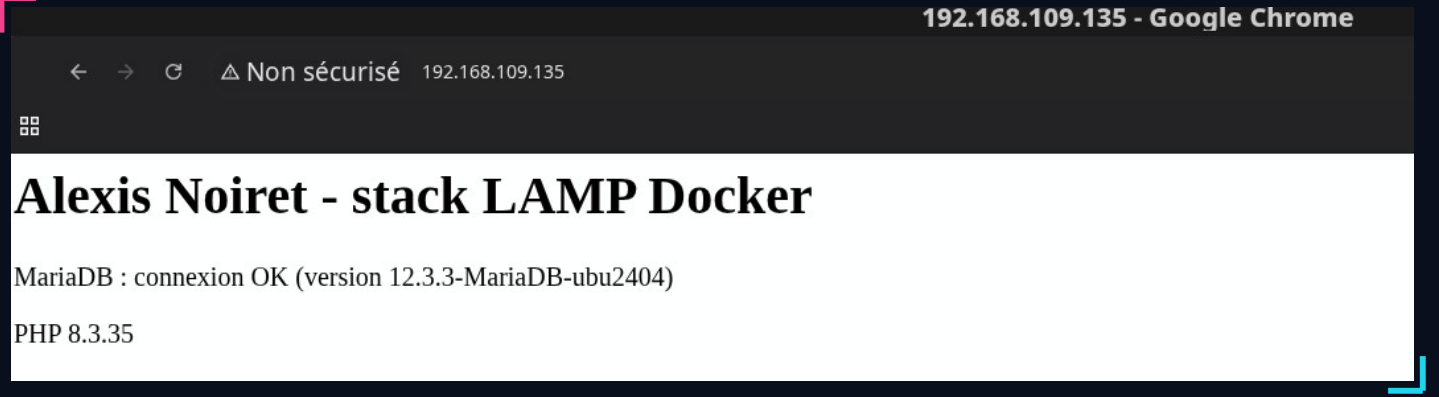
Outil	Type	Points clés	Limites
Dockge	Interface web	Centré Compose, éditeur YAML, terminal web, ~50 Mo de RAM, MIT, maintenu	Pas de Kubernetes ni Swarm
Yacht	Interface web	Modèles d'applications, gestion basique	Abandonné depuis janv. 2023 (v0.0.7-alpha)
Lazydocker	Terminal (TUI)	Un binaire, conteneurs, logs et stats	Pas d'interface graphique
Cockpit	Admin système	Gère tout le serveur, module conteneurs	Surtout orienté Podman
Rancher	Plateforme	Clusters Kubernetes à grande échelle	Surdimensionné pour un hôte

Conclusion Pour un hôte unique sous Compose, Dockge est léger et crédible. Portainer reste plus complet (images, volumes, registries, multi-hôte). Tous montent docker.sock : même risque.

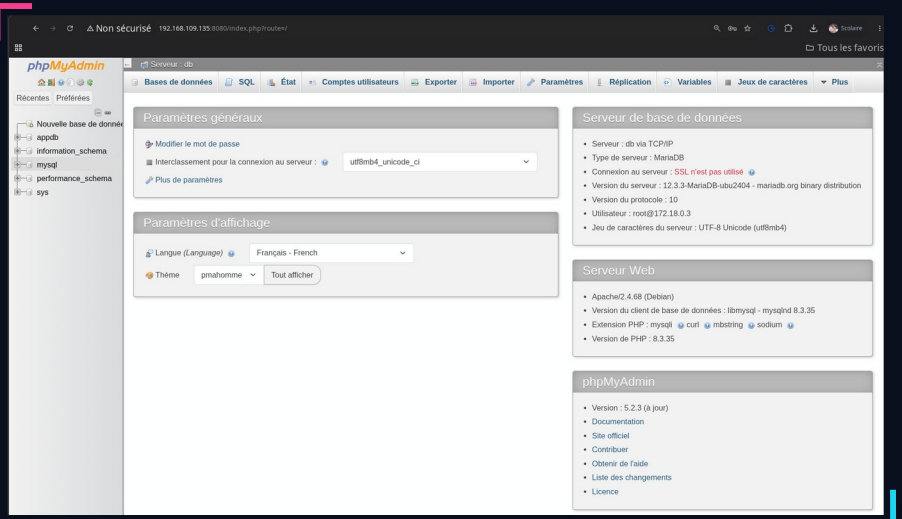
Bonus — Stack LAMP complète

- web : php:8.3-apache + mysqli
- db : mariadb:lts · phpMyAdmin : port 8080
- ftp pour déposer index.php dans le volume web
- La page PHP se connecte à MariaDB

```
$ docker compose up -d --build
$ curl -T index.php ftp://alexis:***@localhost/
```



Bonus — Stack LAMP complète (suite)




```
NAME                IMAGE                          COMMAND                  SERVICE      CREATED          STATUS
      PORTS
bonus-db            mariadb:lts                    "docker-entrypoint.s…"   db           3 seconds ago    Up 2 seconds
      3306/tcp
bonus-ftp           delfer/alpine-ftp-server       "/sbin/tini -- /bin/…"   ftp          3 seconds ago    Up 1 second
      0.0.0.0:21→21/tcp, [::]:21→21/tcp, 0.0.0.0:21000-21010→21000-21010/tcp, [::]:21000-21010→21000-21010/tcp
bonus-phpmyadmin    phpmyadmin:latest              "/docker-entrypoint.…"   phpmyadmin   3 seconds ago    Up 2 seconds
      0.0.0.0:8080→80/tcp, [::]:8080→80/tcp
bonus-web           bonus-web                      "docker-php-entrypoi…"   web          3 seconds ago    Up Less than a se
cond  0.0.0.0:80→80/tcp, [::]:80→80/tcp
```

Bilan et compétences

11 + 1

jobs réalisés + bonus

4

conteneurs dans la stack LAMP

2

scripts install / désinstall

9443

Portainer, interface de gestion

- Images, conteneurs, volumes, réseaux Compose
- Écrire un Dockerfile, publier des ports, partager des données
- Registry privé et interface web
- Automatiser : scripts bash d'installation et de nettoyage
- Sécurité : docker.sock = accès root, mots de passe, ports exposés

DÉPÔT GITHUB

`github.com/alexis-noiret/docker`

README général, un dossier par job avec captures et commandes, scripts de démonstration.

Merci — questions ?